

sTMS Interface Functional Specification

Institute of Technical Education – Student Administration System [Student Records]

Version 1.0

Document Version

Version	Effective Date	Summary of Changes	Created/ Updated By
0.1	10/09/2022	Initial version of ITE Student Administration System IFS	AvePoint
0.2	30/09/2022	Revised specification based on ITE's comments	AvePoint
0.3	04/10/2022	Revised specification based on ITE's comments	AvePoint
0.4	25/10/2022	Revised specification based on ITE's comments	AvePoint
0.5	27/10/2022	Revised specification based on ITE's comments	AvePoint
1.0	28/10/2022	Official version for acceptance and sign-off	AvePoint

Document Acceptance Form

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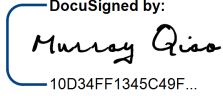
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1. Introduction

1.1 Purpose of this document

This Interface Functional Specification (IFS) is to provide the agreed baseline of communication requirement and detailed technical specification between the SaaS shared Training Management System (sTMS) and ITE Student Administration System. The data will transfer from sTMS to ITE Student Administration System.

sTMS shall identify all ITE's data with changes, generate corresponding CSV files for these data, deposit CSV files encrypted by PGP protocol in the SFTP server accordingly. New scheduled jobs to be implemented by ITE will retrieve, decrypt, and process these PGP encrypted data files accordingly for downstream systems' data needs.

1.2 Intended Audience

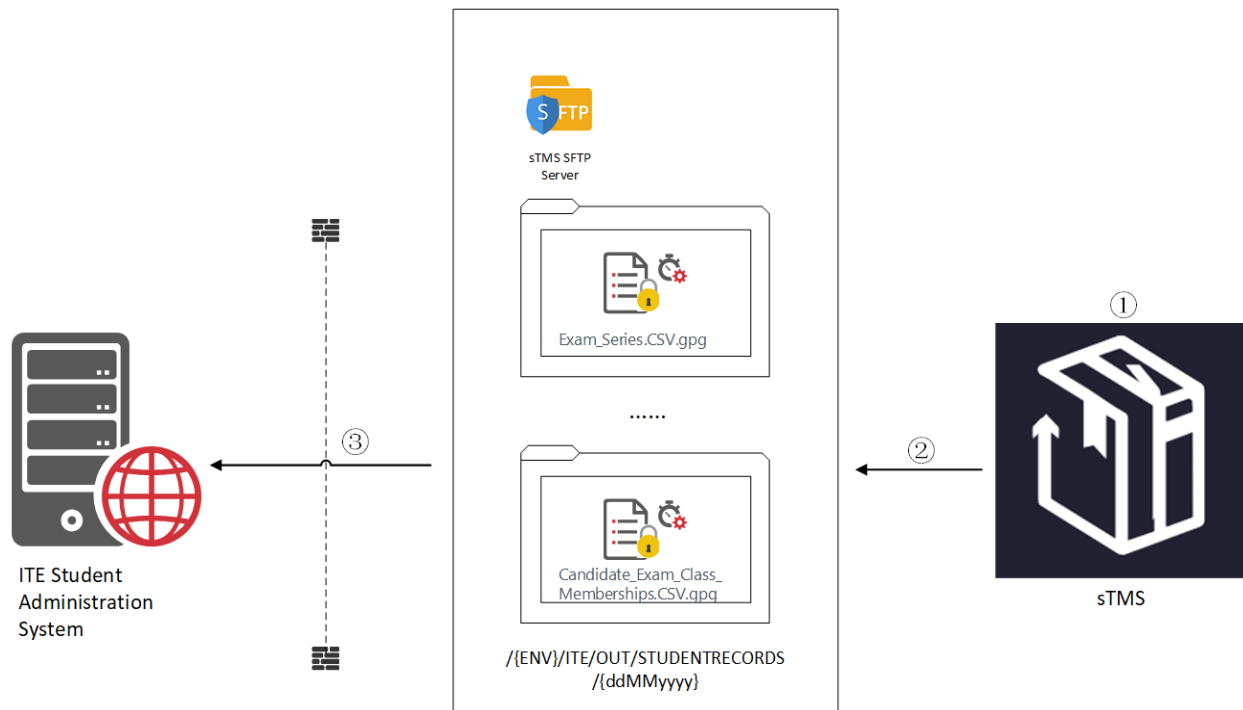
This technical document is used for system communication. Therefore, the intended audiences are the technical stakeholders of both systems.

2. Technical Overview

2.1 System Flow

There are 3 roles involved in the interface diagram.

- **ITE Student Administration System:** Institute of Technical Education Student Administration System
- **sTMS SFTP Server:** A SFTP server hosted in Azure that stores encrypted files used for data exchange.
- **sTMS:** shared Training Management System



*The above system flow only shows some CSV files encryption with PGP protocol. For the interface file name of all ITE Student Administration CSV files, please refer to [2.2 Interface Design Summary](#).

SN	Description
1	sTMS generates ITE CET data CSV files based on institution and modified time on a pre-defined schedule. For file generation rules can refer to 2.3. Business Rule

2	sTMS configures timer job to generate folder /{ENV}/ITE/OUT/STUDENTRECORDS/{ddM-Myyyy} for CSV files. sTMS pushes CSV files to the corresponding folder after encryption with PGP protocol of sTMS SFTP server.
3	ITE Student Administration System pulls these CSV files from sTMS SFTP server on demand.

2.2 Interface Design Summary

Output Interface File

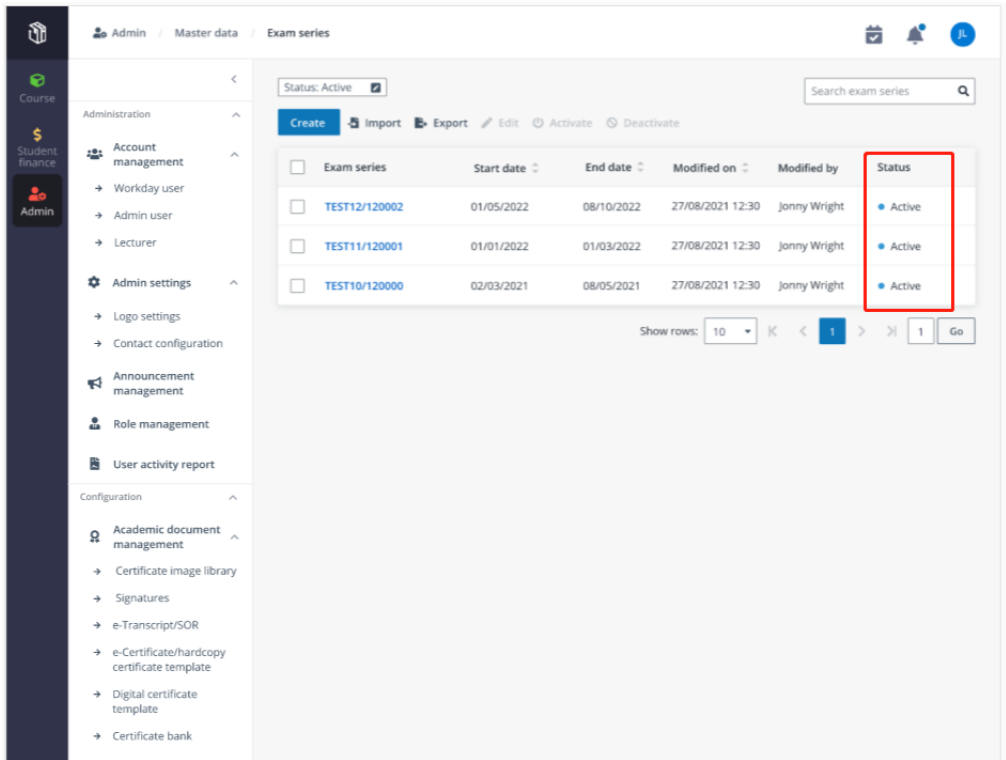
Interface File Name	Exam_Series.CSV.gpg Assessment_Components.CSV.gpg Languages.CSV.gpg Modules.CSV.gpg Campuses.CSV.gpg Exam_Classes.CSV.gpg Candidates.CSV.gpg Candidate_Exam_Class_Memberships.CSV.gpg
Data Source System	shared Training Management System
Data Destination System	ITE Student Administration System
Method	SFTP
File Path	/{ENV}/ITE/OUT/STUDENTRECORDS/{ddMMyyyy} ENV: UAT, PROD
Encryption Method	PGP
Frequency	6 days a week: Monday, Tuesday, Wednesday, Thursday, Friday & Saturday.
Timing	The cut off time for changed data will be 00:00 am every day, timer job trigger timing is next day 01:00 am (TBC)
Folder Retention Policy	Folders under the /{ENV}/ITE/OUT/STUDENTRECORDS are retained for a maximum of 30 days in the sTMS SFTP server, after this period the folders will be automatically deleted. ITE also can delete these folders after pulling out the files

File Transfer Policy	sTMS shall transfer empty files with header info into sTMS SFTP server if there are no new records are added/updated in the table
Possible Exception and Exception Handling	File loss or transfer failure due to sTMS internal exception. E.g., Timer service down, etc. Exception handling steps: • sTMS support team shall be informed by email of the transfer failure. • sTMS support team checks the relevant logs to troubleshoot the cause of the failure. If the issue is due to the transfer files could not be found in the newly created SFTP folder: AP redeposits the files in the newly created corresponding SFTP folder.

2.3 Business Flow

Exam Series Status

Only Exam Series with “Active” Status would be sent. The users would be able to activate or deactivate the exam series under IHL Master Data.

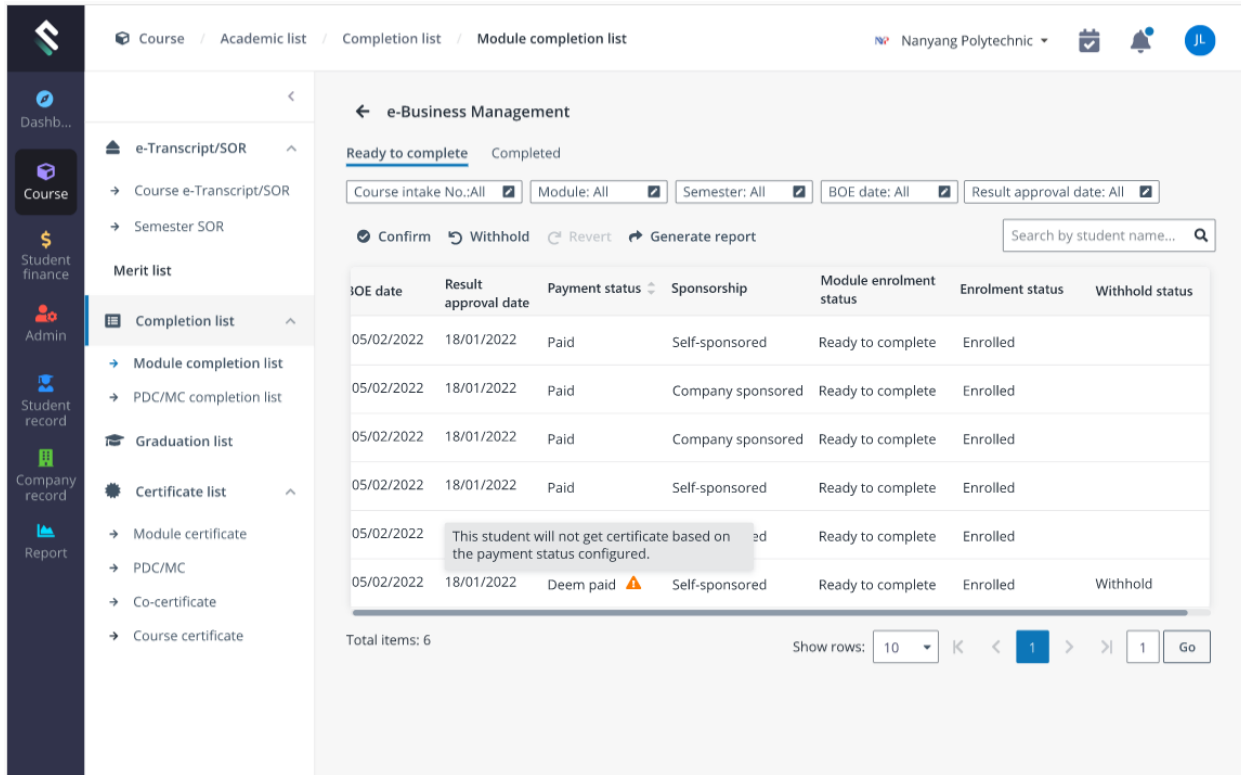


The screenshot displays the 'Exam series' page in the AvePoint Admin interface. The left sidebar contains navigation menus for 'Course', 'Student finance', and 'Admin'. The 'Admin' menu is expanded, showing options like 'Account management', 'Workday user', 'Admin user', 'Lecturer', 'Admin settings', 'Logo settings', 'Contact configuration', 'Announcement management', 'Role management', and 'User activity report'. The main content area shows a table of exam series with columns: Exam series, Start date, End date, Modified on, Modified by, and Status. A red box highlights the 'Status' column, which shows 'Active' for all three entries. The table also includes checkboxes for each row and a 'Show rows' dropdown at the bottom.

Exam series	Start date	End date	Modified on	Modified by	Status
☐ TEST12/120002	01/05/2022	08/10/2022	27/08/2021 12:30	Jonny Wright	Active
☐ TEST11/120001	01/01/2022	01/03/2022	27/08/2021 12:30	Jonny Wright	Active
☐ TEST10/120000	02/03/2021	08/05/2021	27/08/2021 12:30	Jonny Wright	Active

Module Enrolment Status

sTMS shall only send enrolled students' information when Module Enrolment Status is denoted as, In-progress, Make-up exam and/or Ready to Complete.



The screenshot displays the 'e-Business Management' interface for 'Module completion list'. The left sidebar contains navigation links: Dashboard, Course, Student finance, Admin, Student record, Company record, and Report. The main content area shows filters for 'Ready to complete' and 'Completed' status, with dropdowns for Course intake No., Module, Semester, BOE date, and Result approval date. Below the filters are buttons for Confirm, Withhold, Revert, and Generate report, along with a search bar. A table lists student records with columns: BOE date, Result approval date, Payment status, Sponsorship, Module enrolment status, Enrolment status, and Withhold status. The table contains 6 rows of data. A tooltip is visible over the 5th row, stating: 'This student will not get certificate based on the payment status configured.' At the bottom, it shows 'Total items: 6' and a pagination control for 'Show rows: 10'.

BOE date	Result approval date	Payment status	Sponsorship	Module enrolment status	Enrolment status	Withhold status
05/02/2022	18/01/2022	Paid	Self-sponsored	Ready to complete	Enrolled	
05/02/2022	18/01/2022	Paid	Company sponsored	Ready to complete	Enrolled	
05/02/2022	18/01/2022	Paid	Company sponsored	Ready to complete	Enrolled	
05/02/2022	18/01/2022	Paid	Self-sponsored	Ready to complete	Enrolled	
05/02/2022	18/01/2022	Deem paid	Self-sponsored	Ready to complete	Enrolled	Withhold

Flow to generate list of Exam_Classes

SN	Description
1	For a given exam(sub-component), the system locates the corresponding ITE Exam series, module code and subcomponent name.
2	Then sTMS will generate the 'Exam class code' in backend as the key and the value will be "Exam Series_Module Code_Course Intake Number_subcomponent name_default 1_Venue"
3	sTMS checks the exam start date time, facility, facility remarks change daily.
4	sTMS identifies the changes of exam start date time info, and facility ID and facility remarks info in assessment scheduling in CET portal, during the date of checking Delta changes include:

	• Admin edit any of the exam start date time info, and facility ID and facility remarks info in existing schedule data of the sub-component • Admin add any of the exam start date time info, and facility ID and facility remarks info in new schedule data of the sub-component • Admin delete any of the exam start date time info, and facility ID and facility remarks info in existing schedule data • Admin changes any of the facility ID and facility remarks info of IHL master data-Facility which is selected in any assessment schedule info
5	sTMS will send the full load data, generates the output interface file, and deposits it accordingly.

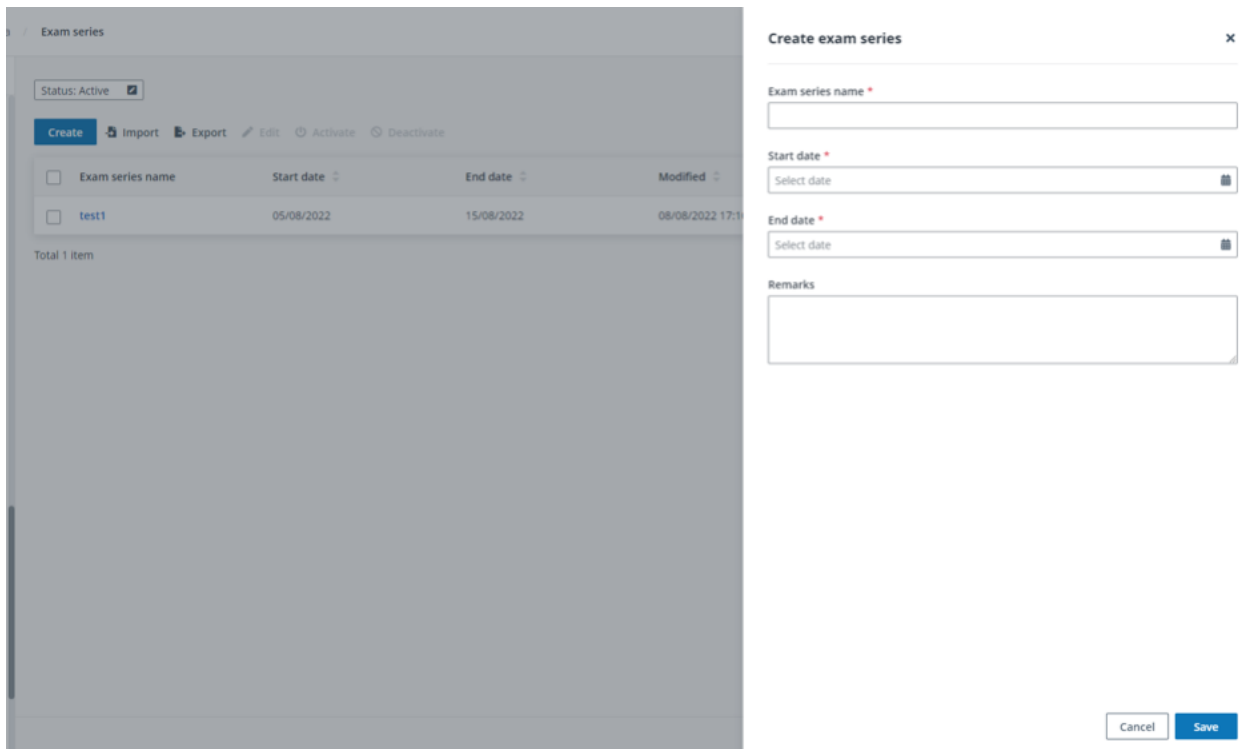
3. Interface Fields

3.1 CSV Files Interface Fields

3.1.1 Entry Requirements Fields for Exam_Series

If exam series is active, and contain modules with online exams, sTMS will send the table records to ITE daily.

IAS Field Name	iStudent Field Name	Field Name in sTMS	Data Type	Remarks
Exam Series	A_EXM_SERIES	Exam Series	CHAR (6)	Display the name of Exam series

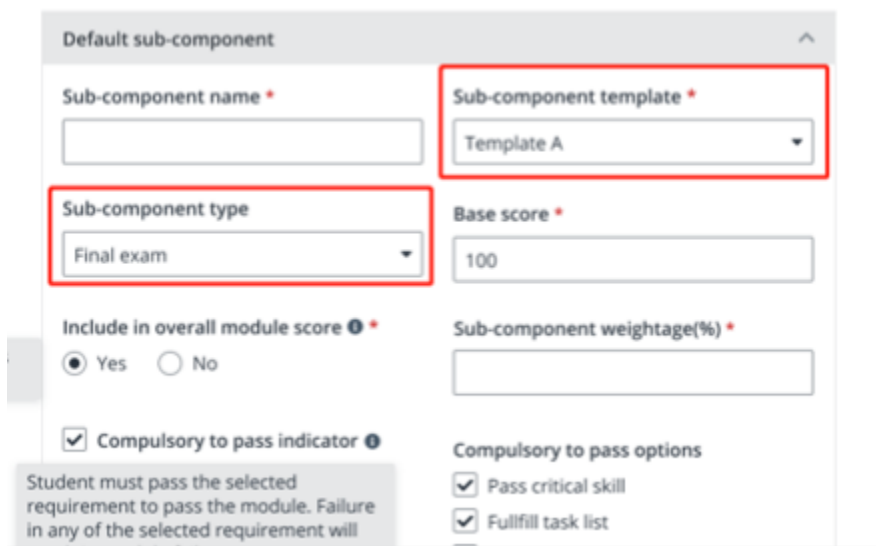


The screenshot displays the 'Exam series' management interface. On the left, a table lists exam series with columns: Exam series name, Start date, End date, and Modified. One item, 'test1', is listed with a start date of 05/08/2022 and an end date of 15/08/2022. Above the table are buttons for 'Create', 'Import', 'Export', 'Edit', 'Activate', and 'Deactivate'. A 'Status: Active' filter is also present. On the right, a 'Create exam series' modal form is open, containing fields for 'Exam series name', 'Start date', 'End date', and 'Remarks'. The 'Start date' and 'End date' fields have 'Select date' buttons. At the bottom right of the modal are 'Cancel' and 'Save' buttons.

3.1.2 Entry Requirements Fields for Assessment_Components

If exam series is active, and contain modules with online exams, sTMS will send the table records to ITE daily.

IAS Field name	iStudent Field Name	Field name in sTMS	Data Type	Remarks
Assessment Component / Assignment Item	A_ASSIGNMENT_ITEM	N/A	CHAR (10)	Default to TEXTM
Exam Series	A_EXM_SERIES	Exam Series	CHAR (6)	Display the name of Exam series



3.1.3 Entry Requirements Fields for Languages

If exam series is active, and contain modules with online exams, sTMS will send the table records to ITE daily.

IAS Field name	iStudent Field Name	Field name in sTMS	Data Type	Remarks
Language	LANGUAGE	N/A	CHAR (10)	Default to English
Exam Series	A_EXM_SERIES	Exam Series	CHAR (6)	Display the name of Exam series

3.1.4 Entry Requirements Fields for Modules

If exam series is active, and contain modules with online exams, sTMS will send the table records to ITE daily.

IAS Field name	iStudent Field Name	Field name in sTMS	Data Type	Remarks
Module code	A_MODULE_CODE	Module code	CHAR (18)	Display the module code. Only send the modules with online test/exams

Exam Series	A_EXM_SERIES	Exam Series	CHAR (6)	Display the name of Exam Series
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3.1.5 Entry Requirements Fields for Campuses

If exam series is active, and contain modules with online exams, sTMS will send the table records to ITE daily.

IAS Field name	iStudent Field Name	Field name in sTMS	Data Type	Remarks
Campus	CAMPUS	Branch	CHAR (5)	Display the branch code. Format: CC-AM, CE-SM, CW-CK. Please refer to Appendix – Branch Mapping for further mapping between sTMS and iStudent Record
Exam Series	A_EXM_SERIES	Exam Series	CHAR (6)	Display the name of Exam Series

3.1.6 Entry Requirements Fields for Exam_Classes

~~sTMS will send Full data of this table based on active Exam series to ITE daily.~~

If exam series is active, and contain modules with online exams, sTMS will send the table records to ITE daily.

IAS Field name	iStudent Field Name	Field name in sTMS	Data Type	Remarks
 Module Code	A_MODULE_CODE	Module code	CHAR (18)	Display the module code
Module Title	COURSE_TITLE_LONG	Module name	CHAR (100)	Display the module name
 Exam class code	A_IAS_EXM_CLASSES_CD	N/A	CHAR (139)	Format: Exam Series_Module Code_Course Intake Number_subcomponent name_default 1_Venue. If no venue yet, venue indicate as X.
Module ID	CRSE_ID	Module ID	CHAR (6)	Use Module code (just the numbers) +0 to 6 digit
Date and time of the exam	A_IAS_EXAM_DTTM	Exam start datetime	CHAR (20)	Display the exam start datetime
Exam Venue	A_EXAM_VENUE	Facility ID	CHAR (10)	Display the facility ID
Exam Venue description	A_EXAM_VENUE_DESCR	Facility name	CHAR (30)	Display the facility Name
Language	LANGUAGE	N/A	CHAR (10)	Default to English
Class Exam Sequence Number	CLASS_EXAM_SEQUENCE	Class exam sequence number	NUMBER	Default to 1

Relay Number	A_EXAM_RELAY	N/A	NUMBER	Default to 0
Class assignment sequence number	SEQUENCE_NO	N/A	NUMBER	Default to 1
Assignment Category	LAM_TYPE	N/A	CHAR (10)	Default to EMTH
Assessment Component/Assignment Item	A_ASSIGNMENT_ITEM	N/A	CHAR (10)	Default to TEXM
Class Number	CLASS_NBR	Class number	NUMBER	Course intake running number (4 digit)
 Exam Series	A_EXM_SERIES	Exam series	CHAR (6)	Display the name of exam series
Term	STRM	N/A	CHAR (4)	Format: TYMM T indicates from TMS Y = last digit of YYYY in course start date. MM = MM of course start date.
Training Scheme	ACAD_CAREER	N/A	CHAR (4)	Default to CET
Session code	SESSION_CODE	N/A	CHAR (3)	Format of exam series to be setup as YYYYMM Last two digit of exam series names.
Academic Organisation	ACAD_ORG	Class name	CHAR (10)	Map to Class Name (10 char)
Campus	CAMPUS	Branch	CHAR (5)	Display the name of Branch, must be in the format of sample value
Action	Action	Action	CHAR (1)	C — Create; U — Update; D — Delete

The symbol  represents the field is the primary key.

← BM0001 - e-Business Management - Final exam

Base score	100
Weightage(%)	20%
Grade mapping rule	Assessment mapping rule 1
Scheduling status	* Draft

+ Add schedule

Schedule name	Facility	Exam date	Start time	End time	Exam duration
Default schedule	Classroom 1	04/05/2022	10:00	11:00	1h
Schedule 1	Classroom 2	04/05/2022	10:00	11:00	1h

Search by student name...

Course : Course intake configuration

Course category: All Course type: All Course start date: All Status: All Search course name or co...

☐	Course intake No.	Course name	Course code	Course category	Course type	Registration duration	Course duration	Assigned role	Status
☐	TP-SD-FQ0001-220001	MSc e-Business Management	TP-SD-FQ0001	Certifiable Course	SkillsFuture Series	01/05/2021 - 08/05/2022	08/05/2021 - 10/08/2022		* Draft
☐	TP-SD-FQ0002-220001	Data Science and Digital Operation	TP-SD-FQ0002	Certifiable Course	SkillsFuture Series	01/05/2021 - 08/05/2022	08/05/2021 - 10/08/2022		Open for application
☐	TP-SD-FQ0003-220001	Cloud and Machine learning	TP-SD-FQ0003	Certifiable Course	SkillsFuture Series	01/05/2021 - 08/05/2022	08/05/2021 - 10/08/2022		Confirmed
☐	TP-SD-FQ0004-220001	Cloud and Machine learning	TP-SD-FQ0004	Certifiable Course	SkillsFuture Series	01/05/2021 - 08/05/2022	08/05/2021 - 10/08/2022		View supporting document
☐	TP-SD-FQ0005-220001	Cloud and Machine learning	TP-SD-FQ0005	Certifiable Course	SkillsFuture Series	01/05/2021 - 08/05/2022	08/05/2021 - 10/08/2022		Cancelled
☒	TP-SD-FQ0006-220001	Cloud and Machine learning	TP-SD-FQ0006	Certifiable Course	SkillsFuture Series	01/05/2021 - 08/05/2022	08/05/2021 - 10/08/2022		Application closed

Show rows: 10 < 1 > 1 Go

3.1.7 Entry Requirements Fields for Candidates

If exam series is active, and contain modules with online exams, sTMS will send the table records to ITE daily.

IAS Field name	iStudent Field Name	Field name in sTMS	Data Type	Remarks
NRIC	EMPLID	UAM Common ID	CHAR (11)	Display UAM Common ID of the student
Complete Name	COMPLETE_NAME	Student Name	CHAR (120)	Display the name of the student
Campus	CAMPUS	Branch	CHAR (5)	Display the name of Branch

				Two scenarios to be considered: a. Students taking more than one exam on the same day, show the first location in sequence: CC->CE->CW b. Students taking more than one exam at different location on different days, first shows the location on the earlier day, then change the location to new location in the interface file on that exam day.
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3.1.8 Entry Requirements Fields for Candidate_Exam_Class_Memberships

If exam series is active, and contain modules with online exams, sTMS will send the table records to ITE daily.

IAS Field name	iStudent Field Name	Field name in sTMS	Data Type	Remarks
NRIC	EMPLID	UAM Common ID	CHAR (11)	Display UAM Common ID of the student
Exam Class Code	A_IAS_EXM_CLASS_CD	N/A	CHAR (139)	Exam Series_Module Code_Course Intake Number_Assessment_subcomponent_default 1_Venue

3.2 Sample Files



Student Records Sample Files.zip

4. Constraints

No.	Constraints
1	ITE would be creating module tag in sTMS to indicate online test/exams.
2	Only the data has the relationship to online tests/exams under the module should be sent over to ITE

5. Appendix – Branch Mapping

Branch mapping from ITE to sTMS

ITE Campus	ITE Branch Code in sTMS	ITE Branch Name in sTMS
CC-AM	CC	ITE College Central
CE-SM	CE	ITE College East
CW-CK	CW	ITE College West